

A constant-one finite-determination subcase for classical L_p spaces

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Source Question

The source paper is Sofiya Ostrovska and Mikhail I. Ostrovskii, *Distortion in the finite determination result for embeddings of locally finite metric spaces into Banach spaces*, arXiv:1510.05974. It defines $D(X) \leq \alpha$ to mean that whenever all finite subsets of a locally finite metric space M embed into X with distortion at most C , the whole space M embeds into X with distortion at most αC . Its final problem asks:

Problem 5.1. Do there exist Banach spaces X with $D(X) > 1^+$?

the result stated in Theorem 1.14 is proved.

□

5 An open problem

In our opinion the most interesting open problem related to this study is:

Problem 5.1. *Do there exist Banach spaces X with $D(X) > 1^+$?*

Result

Theorem 1. *For every $1 \leq p < \infty$,*

$$D(L_p[0, 1]) = 1.$$

Equivalently, if M is locally finite and all finite subsets of M embed into $L_p[0, 1]$ with distortion at most C , then M itself embeds into $L_p[0, 1]$ with distortion at most C .

This is a partial negative answer to the source problem: the classical nonatomic spaces $L_p[0, 1]$, $1 \leq p < \infty$, cannot witness a strict finite-determination gap. The existence of some other Banach space X with $D(X) > 1^+$ remains open.

Idea of the Proof

The usual finite-determination compactness argument first embeds M into an ultrapower of $L_p[0, 1]$ with the same distortion C . The point is that, for classical $L_p[0, 1]$, this ultrapower detour loses no metric information: ultrapowers of L_p -spaces are again L_p -spaces, and the countable image of a locally finite metric space spans only a separable subspace. That separable subspace is contained in a separable L_p -space, and every separable L_p -space over a finite measure algebra embeds linearly isometrically into $L_p[0, 1]$. Composing the ultrapower embedding with this linear isometry returns the metric copy to $L_p[0, 1]$ without changing the distortion.

Proof

Let $X = L_p[0, 1]$, $1 \leq p < \infty$. Let M be locally finite and suppose that all finite subsets of M admit bilipschitz embeddings into X with distortion at most C . Fix a base point $o \in M$. Since M is locally finite, $M = \bigcup_{n \geq 1} B(o, n)$ is countable.

Let $\mathcal{F}$ be the directed set of finite subsets of M containing o , and choose an ultrafilter $\mathcal{U}$ on $\mathcal{F}$ containing every tail

$$\{F \in \mathcal{F} : E \subseteq F\},$$

where $E \subseteq M$ is finite. For each $F \in \mathcal{F}$, choose an embedding $f_F : F \rightarrow X$, translate it so that $f_F(o) = 0$, and rescale it so that

$$d(u, v) \leq \|f_F(u) - f_F(v)\|_X \leq Cd(u, v) \quad (u, v \in F).$$

For $x \notin F$, set $f_F(x) = 0$. For each fixed $x \in M$, the family $(f_F(x))_{F \in \mathcal{F}}$ is bounded, because on the $\mathcal{U}$ -large set $\{F : x \in F\}$ its norm is at most $Cd(x, o)$, and elsewhere it is zero. Hence

$$f(x) = [(f_F(x))_{F \in \mathcal{F}}]_{\mathcal{U}}$$

defines a map $f : M \rightarrow X^{\mathcal{U}}$. For any $x, y \in M$, the set of indices F containing $\{o, x, y\}$ is $\mathcal{U}$ -large, so

$$d(x, y) \leq \|f(x) - f(y)\|_{X^{\mathcal{U}}} \leq Cd(x, y). \quad (1)$$

It remains to bring this countable subset of $X^{\mathcal{U}}$ back into X isometrically. By the standard Kakutani–Heinrich representation theorem for ultraproducts of L_p -spaces, $X^{\mathcal{U}}$ is itself an L_p -space over a finite measure algebra. Let

$$E = \overline{\text{span}} f(M) \subseteq X^{\mathcal{U}}.$$

This is separable. In an $L_p(\Omega, \Sigma, \mu)$ representation of $X^{\mathcal{U}}$, choose a countable dense subset of E . The sigma-algebra generated by these countably many functions is countably generated, and the corresponding closed sublattice $L_p(\Sigma_0)$ is a separable L_p -space containing E .

Every separable L_p -space over a finite measure algebra decomposes as the p -sum of a nonatomic part and at most countably many atoms. Such a space embeds linearly isometrically into $L_p[0, 1]$: realize the nonatomic part on a subinterval and realize the countably many atoms on pairwise disjoint intervals of the appropriate measures. Thus there is a linear isometry

$$J : L_p(\Sigma_0) \longrightarrow L_p[0, 1].$$

Composing f with J and using (1) gives

$$d(x, y) \leq \|Jf(x) - Jf(y)\|_{L_p[0, 1]} \leq Cd(x, y) \quad (x, y \in M).$$

Therefore M embeds into $L_p[0, 1]$ with distortion at most C . Since the definition has $\alpha \geq 1$, this proves $D(L_p[0, 1]) = 1$.

Verification and Limitations

The proof uses only standard facts about L_p -spaces: ultraproducts of L_p -spaces are L_p -spaces, and separable finite-measure L_p -spaces embed linearly isometrically into $L_p[0, 1]$. No computational verification is involved.

This does not solve Problem 5.1 in full. It rules out the classical nonatomic $L_p[0, 1]$ spaces as witnesses for a strict gap, but it does not address arbitrary Banach spaces, nor spaces such as ℓ_p where the source paper already proves $D \leq 1^+$ by a different argument.

Novelty Check

The local run indexes contained a previous unfinished attempt for arXiv:1510.05974 but no packet for this $L_p[0, 1]$ subcase. Searches of the local source corpus and web queries combining $D(X) > 1^+$, L_p , finite determination, and locally finite metric spaces did not reveal an explicit later answer. Catrina–Ostrovskii’s 2023 paper improves the general finite-determination constant to $3 + \varepsilon$ and restates the $1 + \varepsilon$ problem, but does not record the constant-one pullback for classical $L_p[0, 1]$.

References

References

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- [4] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces II: Function Spaces*, Springer, 1979.